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United States Patent	12416774
Kind Code	B2
Date of Patent	September 16, 2025
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Method of manufacturing lens and light emitting device

Abstract

A method of manufacturing a lens includes: injecting a thermosetting first resin in a first mold, and curing the thermosetting first resin, to form a cover blank having cover parts; removing a part or all parts of the first mold; arranging the cover blank in a second mold; injecting a thermosetting second resin having a greater light absorptance or a greater light reflectance than the thermosetting first resin into the second mold and curing the thermosetting second resin, to form a lens blank having a light-shielding part between adjacent ones of the cover parts; taking out the lens blank from the second mold; cutting the lens blank at the light-shielding part located between the adjacent ones of the cover parts to obtain individual lenses having lateral side walls and flange parts extending outward from lower-end portions of the lateral side walls, which are both covered by the light-shielding part.

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Appl. No.: 18/075375

Filed: December 05, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230102235 A1	Mar. 30, 2023

Foreign Application Priority Data

JP	2018-162680	Aug. 31, 2018
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Related U.S. Application Data

division parent-doc US 16558275 20190902 US 11640038 child-doc US 18075375

Publication Classification

Int. Cl.: G02B7/02 (20210101); B29B11/08 (20060101); B29D11/00 (20060101); G02B7/00 (20210101); H10H20/01 (20250101); H10H20/814 (20250101)

U.S. Cl.:

CPC G02B7/02 (20130101); B29B11/08 (20130101); B29D11/00269 (20130101); G02B7/003 (20130101); H10H20/034 (20250101); H10H20/814 (20250101)

Field of Classification Search

CPC: G02B (7/02); B29B (11/08); B29D (11/00269)

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATION (1) The present application is a divisional application of the U.S. patent application Ser. No. 16/558,275 filed on Sep. 2, 2019, which claims priority under 35 U. S. C. § 119 to Japanese Patent Application No. 2018-162680, filed Aug. 31, 2018. The contents of Japanese Patent Application No. 2018-162680 and the U.S. patent application Ser. No. 16/558,275 are incorporated herein by reference in its entirety.

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present disclosure relates to a lens and a light emitting device, and a method of manufacturing the lens and the light emitting device.

Discussion of the Background

(2) Applications for flush light sources used in cellular phones and other mobile devices, a light emitting device may include an LED element mounted on a circuit board, a cover having an optical lens disposed facing the LED element, and a light-reflecting member integrally formed with the cover, for example, described in Japanese patent publication No. 5139915. In the light emitting device described in Japanese patent publication No. 5139915, a metal film is used as the light-reflecting member and the metal film is formed by way of, for example, vacuum vapor deposition.

SUMMARY OF THE INVENTION

(3) A lens according to one embodiment of the present disclosure includes a cover part and a light-shielding part. The cover part includes: a lens part having one or more lateral sides; a connection part constituting one or more lateral side walls each extending downward from a respective one of the one or more lateral sides of the lens part; and one or more flange parts each extending outward from a lower-end portion of a corresponding one of the one or more lateral side walls. The lens part and the connection part define a recess having an opening facing downward. The lens part defines a bottom surface of the recess. The one or more lateral side walls define lateral surfaces of the recess, which define the opening of the recess. The one or more flange parts extend outward from a periphery of the opening of the recess. The lens part, the one or more flange parts, and the connection part are formed of a thermosetting first resin and continuous to one another. The light-shielding part covers outer lateral surfaces of the one or more lateral side walls and are formed of a thermosetting second resin having a greater light-absorptance or a greater light-reflectance than the thermosetting first resin. The flange part has a greater thickness than the connection part.

(4) A lens according to one embodiment of the present disclosure includes a cover part and a light-shielding part. The cover part includes: a lens part having one or more lateral sides; a connection part constituting one or more lateral side walls each extending downward from a respective one of the one or more lateral sides of the lens part; and one or more flange parts each extending outward from a lower-end portion of a corresponding one of the one or more lateral side walls. The lens part, the one or more flange parts, and the connection part are formed of a thermosetting first resin and are continuous to one another. The one or more flange parts each have a thickness greater than 30 μm . The light-shielding part covers outer lateral surfaces of the one or more lateral side walls and is formed of a thermosetting second resin having a greater light-absorptance or a greater light-reflectance than the thermosetting first resin.

(5) A light emitting device according to one embodiment of the present disclosure includes a light emitting element and a lens. The lens includes a cover part and a light-shielding part. The cover part includes: a lens part having one or more lateral sides; a connection part constituting one or more lateral side walls each extending downward from a respective one of the one or more lateral sides of the lens part; and one or more flange parts each extending outward from a lower-end portion of a corresponding one of the one or more lateral side walls. The lens part and the connection part define a recess having an opening facing downward. The lens part defines a bottom surface of the recess. The one or more lateral side walls define lateral surfaces of the recess, which define the opening of the recess. The one or more flange parts extend outward from a periphery of the opening of the recess. The lens part, the one or more flange parts, and the connection part are formed of a thermosetting first resin and are continuous to one another. The light-shielding part covers outer lateral surfaces of the one or more lateral side walls and is formed of a thermosetting second resin having a greater light-absorptance or a greater light-reflectance than the thermosetting first resin. The one or more flange parts have a greater thickness than the connection part. The lens part is disposed at a location allowing light from the light emitting element to be transmitted through the lens part.

(6) A light emitting device according to one embodiment of the present disclosure includes a light emitting element and a lens. The lens includes a cover part and a light-shielding part. The cover part includes: a lens part having one or more lateral sides; a connection part constituting one or more lateral side walls each extending from a respective one of the one or more lateral sides of the

lens part; and one or more flange parts each extending outward from a lower-end portion of a corresponding one of the one or more lateral side walls. The lens part, the one or more flange parts, and the connection part are formed of a thermosetting first resin and are continuous to one another, with a thickness of each of the one or more flange parts being greater than 30 μm . The light-shielding part covers outer lateral surfaces of the one or more lateral side walls and is formed of a thermosetting second resin having a greater light-absorptance or a greater light-reflectance than the thermosetting first resin. The lens part is disposed at a location allowing light from the light emitting element to be transmitted through the lens part.

(7) A method of manufacturing a lens according to one embodiment of the present disclosure includes: forming a cover blank, the forming a cover blank including, injecting a thermosetting first resin in a first mold and curing the first resin, to form a cover blank having a plurality of cover parts, each of the cover parts including a lens part having one or more lateral sides, a connection part constituting one or more lateral side walls each extending downward from a respective one of the one or more lateral sides of the lens part, and one or more flange parts each extending outward from a lower-end portion of a corresponding one of the one or more lateral side walls, the lens part, the flange part, and the connection part being continuous to one another, the lens part and the connection part defining a recess having an opening facing downward, the one or more flange parts extending outward from a periphery of the opening of the recess; removing a part or all parts of the first mold; arranging the cover blank in a second mold; forming a lens blank, the forming a lens blank including, injecting a thermosetting second resin having a greater light absorptance or a greater light reflectance than the thermosetting first resin into the second mold, and curing the thermosetting second resin, to form a lens blank having a light-shielding part between adjacent ones of the cover parts; and obtaining individual lenses, the obtaining individual lenses including, taking out the lens blank from the second mold, and cutting the lens blank at the light-shielding part located between adjacent ones of the cover parts to obtain individual lenses each with an entirety of outer surfaces of the one or more lateral side walls covered by the light-shielding part and each with an entirety of upper surfaces of the one or more flange parts covered by the light-shielding part.

(8) A method of manufacturing a lens according to one embodiment of the present disclosure includes: forming a cover blank, removing a part or an entire of the first mold, arranging the cover blank in a second mold, forming a lens blank, and obtaining individual lenses. The forming a cover blank includes injecting a thermosetting first resin in a first mold and curing the first resin to form a cover blank having a plurality of cover parts, each of the cover parts including a lens part having one or more lateral sides, a connection part constituting one or more lateral side walls each extending downward from a respective one of the one or more lateral sides of the lens part, and one or more flange parts each extending outward from a lower-end portion of a corresponding one of the one or more lateral side walls. The lens part, the flange part, and the connection part being continuous to one another. The one or more flange parts have a thickness greater than a thickness of the connection part. The lens part and the connection part define a recess having an opening facing downward. The lens part defines a bottom surface of the recess. The one or more lateral side walls constituted by the connection part define lateral surface(s) of the recess, which define the opening of the recess. The recess is located inward of the one or more flange parts. The removing a part or an entire of the first mold is performed after the cover blank is formed. In the arranging the cover mold in a second mold, the cover blank is arranged in a second mold. The forming a lens blank includes injecting a thermosetting second resin in the second mold, the second resin having a greater light absorptance or a greater light reflectance than the thermosetting first resin, and curing the second resin to form a light-shielding part between adjacent ones of the cover parts. The obtaining individual lenses includes taking out the lens blank from the second mold and cutting the lens blank at the light-shielding part located between the adjacent ones of the cover parts to obtain individual lenses each having the outer lateral sides of the connection part (and an upper surface of

each of the one or more flange parts) covered by the light-shielding part.

(9) A method of manufacturing a lens according to one embodiment of the present disclosure includes: forming a cover blank, the forming a cover blank including, injecting a thermosetting first resin in a first mold, and curing the first resin, to form a cover blank having a plurality of cover parts, each of the cover parts including a lens part having one or more lateral sides, a connection part constituting one or more lateral side walls each extending downward from a respective one of the one or more lateral sides of the lens part, and one or more flange parts each extending outward from a lower-end portion of a corresponding one of the one or more lateral side walls, the lens part, the one or more flange parts, and the connection part being continuous to one another, the one or more flange parts having a thickness greater than 30 μm ; removing a part or all parts of the first mold; arranging the cover blank in a second mold; forming a lens blank, the forming a lens blank including, injecting a thermosetting second resin having a greater light absorptance or a greater light reflectance than the thermosetting first resin into the second mold and curing the thermosetting second resin, to form a lens blank having a light-shielding part between adjacent ones of the cover parts; and obtaining individual lenses, the obtaining individual lenses including, taking out the lens blank from the second mold, and cutting the lens blank at the light-shielding part located between the adjacent ones of the cover parts to obtain individual lenses each with an entirety of outer surfaces of the one or more lateral side walls covered by the light-shielding part and each with an entirety of upper surfaces of the one or more flange parts covered by the light-shielding part.

(10) A method of manufacturing a lens according to one embodiment of the present disclosure includes forming a cover blank, the forming a cover blank including, injecting a thermosetting first resin in a first mold, and curing the thermosetting first resin, to form a cover blank having a plurality of cover parts, each of the cover parts including a lens part having one or more lateral sides, a connection part constituting one or more lateral side walls each extending downward from a respective one of the one or more lateral sides of the lens part, and one or more flange parts each extending from a lower-end portion of a corresponding one of the one or more lateral side walls, each of the one or more flange parts having a greater thickness than the connection part, and the lens part, the one or more flange parts, and the connection part being continuous to one another; removing the first mold; cutting the cover blank at the flange part located between adjacent ones of the cover parts; arranging the cut cover blank in a second mold; forming a lens blank, the forming the lens blank including, injecting a thermosetting second resin having a greater light absorptance or a greater light reflectance than the thermosetting first resin into the second mold and curing the thermosetting second resin to form a lens blank having a light-shielding part located between adjacent ones of the cover parts; and obtaining individual lenses, the obtaining individual lenses including, taking out the lens blank from the second mold, cutting the lens blank at the light-shielding part located between the adjacent ones of the cover parts to obtain individual lenses each with an entirety of outer surfaces of the one or more side walls covered by the light-shielding part, each with an entirety of upper surfaces of the one or more flange parts covered by the light-shielding part, and each with an outer end surface of each of the one or more flange parts covered by the light-shielding part.

(11) A method of manufacturing a light emitting device according to one embodiment of the present disclosure includes providing a lens according one of the methods described above, and arranging the lens such that light from the light emitting element is transmitted through the lens part of the lens.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) A more complete appreciation of the invention and many of the attendant advantages thereof will be readily obtained as the same becomes better understood by reference to the following detailed description when considered in connection with the accompanying drawings, wherein:
- (2) FIG. 1 is a perspective sectional view schematically showing a structure of a light emitting device that includes a lens according to a first embodiment, taken along a line passing through the center of the light emitting device;
- (3) FIG. 2 is a plan view schematically showing a structure of a lens according to the first embodiment;
- (4) FIG. 3 is a cross-sectional view, taken along line III-III of FIG. 2, schematically showing a structure of a lens according to the first embodiment, in which the line is passing through the center of the lens
- (5) FIG. 4 is a cross-sectional view schematically showing a structure of a light emitting device according to the first embodiment, taken along a line passing through the center of the light emitting device;
- (6) FIG. 5 is a flow chart showing a procedure of a method of manufacturing a lens according to the first embodiment;
- (7) FIG. 6A is a cross-sectional view schematically illustrating forming a cover blank in a method of manufacturing a lens according to the first embodiment;
- (8) FIG. 6B is a cross-sectional view schematically showing removing a first mold in a method of manufacturing a lens according to the first embodiment;
- (9) FIG. 6C is a cross-sectional view schematically illustrating arranging a cover blank in a second mold and forming a lens blank in a method of manufacturing a lens according to the first embodiment;
- (10) FIG. 6D is a cross-sectional view schematically illustrating cutting the lens blank to obtain individual lenses in a method of manufacturing a lens according to the first embodiment;
- (11) FIG. 7A is a cross-sectional diagram illustrating a state of arranging a lens according to the first embodiment via an engaging member with respect to a substrate having a light emitting element mounted thereon;
- (12) FIG. 7B is a cross-sectional diagram illustrating a state of engaging a flange part of a lens according to the first embodiment with an engaging member;
- (13) FIG. 7C is a cross-sectional view illustrating a state the lens according to the first embodiment disposed on the substrate, with the flange part of the lens engaged with the engaging member;
- (14) FIG. 8A is a cross-sectional view schematically showing a structure of a light emitting device that includes a lens according to a second embodiment, taken along a line passing through the center of the light emitting device;
- (15) FIG. 8B is a cross-sectional view schematically illustrating arranging a cover blank in a second mold to form a lens blank in a method of manufacturing a lens according to the second embodiment;
- (16) FIG. 8C is a cross-sectional view schematically illustrating cutting the lens blank to obtain individual lenses in a method of manufacturing a lens according to the second embodiment;
- (17) FIG. 9 is a cross-sectional view schematically showing a structure of a light emitting device that includes a lens according to a third embodiment, taken along a line passing through the center of the light emitting device;
- (18) FIG. 10 is a flow chart showing a procedure of a method of manufacturing a lens according to the third embodiment;
- (19) FIG. 11A is a cross-sectional view schematically illustrating forming a cover blank in a method of manufacturing a lens according to the third embodiment;
- (20) FIG. 11B is a cross-sectional view schematically illustrating cutting a cover blank in a method

of manufacturing a lens according to the third embodiment;

(21) FIG. 11C is a cross-sectional view schematically illustrating arranging a cut cover blank in a second mold and forming a lens blank in a method of manufacturing a lens according to the third embodiment;

(22) FIG. 11D is a cross-sectional view schematically illustrating cutting a lens blank to singulate lenses in a method of manufacturing a lens according to the third embodiment;

(23) FIG. 12A is a plan view schematically showing a structure of a lens according to a fourth embodiment;

(24) FIG. 12B is a schematic perspective cross-sectional view taken along line XIIB-VIIB of FIG. 12A;

(25) FIG. 13A is a plan view schematically showing a structure of a lens according to a fifth embodiment;

(26) FIG. 13B is a schematic perspective cross-sectional view taken along line XIIIIB-VIIIB of FIG. 13A;

(27) FIG. 14 is a plan view schematically showing a structure of a lens according to a sixth embodiment;

(28) FIG. 15 is a cross-sectional view schematically showing a structure of a light emitting device according to a seventh embodiment, taken along a line passing through the center of the light emitting device;

(29) FIG. 16 is a cross-sectional view schematically showing a structure of a light emitting device according to an eighth embodiment;

(30) FIG. 17A is a schematic plan diagram showing a variational example of a flange part of a lens and illustrating an engaged state of the flange part with an engaging part;

(31) FIG. 17B is a schematic plan diagram showing another variational example of a flange part of a lens and illustrating an engaged state of the flange part with an engaging part; and

(32) FIG. 17C is a schematic plan diagram showing a variational example of a lens and illustrating an engaged state of the lens with an engaging part.

DESCRIPTION OF THE EMBODIMENTS

(33) In the following, certain embodiments will be described with reference to the drawings. However, the embodiments shown below are exemplify lenses, light emitting devices for embodying the technical idea of the present invention, and the technical idea of the present invention is not limited thereto. The sizes, materials, shapes and the relative positions of the members described in the embodiments are given as examples and not as a limitation to the scope of the invention unless specifically stated. The sizes and positional relationships of the members in each of drawings are occasionally shown exaggerated for ease of explanation.

First Embodiment

(34) FIG. 1 is a perspective sectional view schematically showing a structure of a light emitting device that includes a lens according to a first embodiment, taken along a line passing through the center of the light emitting device. FIG. 2 is a plan view schematically showing a structure of a lens according to the first embodiment. FIG. 3 is a cross-sectional view, taken along line III-III of FIG. 2, schematically showing a structure of a lens according to the first embodiment, in which the line is passing through the center of the lens. FIG. 4 is a cross-sectional view schematically showing a structure of a light emitting device according to the first embodiment, taken along a line passing through the center of the light emitting device.

(35) The light emitting device **100** includes a light emitting element **51** and a lens **51**. The light emitting device **100** further includes a base **52** on which the light emitting element **51** is mounted.

Lens

(36) Lens **1** will be described.

(37) The lens **1** includes a cover part **2** and a light-shielding part **3**.

(38) The cover part **2** of the lens **1** includes a lens part **21** having one or more lateral sides, a

connection part **22** constituting one or more lateral side walls each extending downward from a respective one of the one or more lateral sides of the lens part **21**, and a flange part **23** extending outward from a lower-end portion of a corresponding one of the lateral side walls constituted by the connection part **22**. The lens part **21** and the connection part **22** are formed of a thermosetting first resin and continuous to one another. The light-shielding part **3** of the lens **1** covers outer lateral surfaces of the lateral side walls constituted by the connection part **22** (and an upper surface of the flange part) and formed of a thermosetting second resin having a greater light-absorptance or a greater light-reflectance than the thermosetting first resin. The lens **1** is, for example, used as a flash lens for a cellular phone or the like.

Cover Part

(39) The cover part **2** includes a lens part **21**, a flange part **23**, and a connection part **22**, which are formed of a thermosetting first resin and continuous to one another. In the cover part **2**, the lens part **21** and the connection part **22** define a recess **4** having an opening facing downward, the lens part defines a bottom surface of the recess, and the connection part defines lateral surfaces of the recess. The lateral surfaces of the recess defines the opening of the recess. The flange part **23** extends outward from a periphery of the opening of the recess **4**.

(40) The lens part **21** is a member through which light from the light emitting element **51** can be emitted to the outside in parallel light, condensed light, or diffused light. The lens part **21** includes a light emitting surface **21a** where light emitted from the light emitting element **51** enters, and a light emitting surface **21b** at an opposite side to the light incidence surface **21a** where the incident light is refracted and is emitted to the outside.

(41) The outer shape in a plan view of the lens part **21** can be selected from various appropriate shapes, examples thereof include a polygonal shape such as a quadrangular shape, a hexagonal shape, and an octagonal shape, a circular shape and an elliptic shape. A center portion of the lens part **21** serves as a lens preferably has a circular shape or an elliptic shape, where a circular shape is more preferable. The lens part **21** has a maximum thickness of, for example, in a range of 0.1 mm to 10 mm, preferably in a range of 0.5 mm to 5 mm.

(42) For the lens part **21**, a Fresnel lens, a total internal reflection (TIR) lens, or the like can be used, in which, a Fresnel lens is preferable. The Fresnel lens **21** may have either a single lens-center or a plurality of lens-centers. When a plurality of Fresnel lenses **21** are employed, the Fresnel lenses are arranged corresponding to the arrangement of the light emitting element **51**, such that an odd number of Fresnel lenses are preferably arranged in a zigzag form, and an even number of Fresnel lenses are preferably arranged in a square matrix form.

(43) The Fresnel lens **21** includes a plurality of concentric circles of ridges **21c** on the light incidence surface **21a** and on the light emitting surface **21b**. When combined, the cross-sectional shapes of the plurality of ridges **21c** form a curved surface of a single convex lens. The plurality of ridges **21c** are preferably arranged in concentric circles or concentric ellipsoids in a radial direction of the Fresnel lens **21**. In the Fresnel lens **21**, the base plane of the plurality of ridges **21c** can be flat, or concave or convex.

(44) Each of the ridges **21c** has a cross-sectional shape formed with a straight portion at the center-side and a segment of a curved surface of a lens at the outer-side. The profile of each of the ridges **21c** may be formed with inwardly curved segment either a concave curve or a convex curve, in conformity to the direction of light to be emitted. The angles (Fresnel angles) at the tips of the ridges **21c** are adjusted such that light from the light emitting element **51** is emitted to the outside in parallel light.

(45) The connection part **22** constitutes lateral side walls each extending downward from a respective one of the lateral sides of the lens part **21** such that an upper end portion of each side wall is contiguous to an upper end portion of the respective one of the lateral sides of the lens part **21**, that the side walls constituted by the connection part **22** extend downward in a right angle with respect to the lens part **21**, and that a lower-end portion of the side walls of the connection part **22**

is contiguous to the flange part **23** in, for example, a right angle. The connection part **22** and the lens part **21** define the recess **4** having the opening facing downward to accommodate a light emitting element **51**. In a cross-sectional view, the recess **4** is preferably of a rectangular U-shape, but a semicircular shape or a semi-elliptical shape can also be employed. The connection part **22** is preferably formed such that the lens part **21** is located at a center of a bottom surface of the recess **4**. The lens part **21** has a rectangular outer peripheral shape in a plan view and the connection part **22** is disposed with respect to each side of the rectangular shape, with the opposite sides of the connection part **22** being in parallel to each other. The connection part **22** has a thickness smaller than a thickness of the flange part **23**. When the connection part **22** is formed with a thickness smaller than a thickness of the flange part **23**, the thickness of the connection part **22** can be, for example, preferably in a range of 30 μm to 100 μm , more preferably in a range of 30 μm to 200 μm .

(46) The flange part **23** is formed continuous to and protruding outward from and perpendicular to a lower end portion of corresponding one of the lateral side walls constituted by the connection part **22**. The flange part **23** has a thickness greater than a thickness of the connection part **22**. The flange part **23** has a frame-like shape and formed along the lateral side walls of the connection part **22** such that the flange part **23** entirely surrounds a lower-end portion of an outer periphery of the lateral side walls constituted by the connection part **22** as shown in FIGS. **1** and **2**, and is used for bonding or securing to a substrate **52** on which the light emitting element **51** is mounted.

Incidentally, note that the flange part **23** may be constituted by one or more separate parts extending from and partially surrounding the lower-end portion of the outer periphery of the lateral side walls constituted by the connection part **22**. That is, each of the one or more separate parts of the flange part **23** may be formed at least along a portion of the lower-end portion of the outer periphery of the lateral side walls constituted by the connection part **22**.

(47) More specifically, the flange part **23** has a thickness (**D2**) greater than a thickness (**D1**) of the connection part **22** ($\text{D1} < \text{D2}$) and also greater than 30 μm . The flange part **23** has a thickness in a range greater than 30 μm to 1000 μm , preferably in a range of 50 to 1000 μm or, more preferably in a range of 100 μm to 1000 μm , further preferably in a range greater than 200 μm to 1000 μm . Thus, the flange part **23** has a thickness greater than the thickness of the connection part **22**, which allows for stable connection of the lens **1** to an engaging part **54** (for example, shown in FIG. **7A**).

(48) The length of the flange part **23** protruding outward from the lower end portion of the lateral side walls constituted by the connection part **22** is preferably in a range of 200 μm to 3000 μm , more preferably in a range of 200 μm to 1000 μm . With this arrangement, stable bonding or securing between the flange part **23** and the substrate **52** can be obtained. This arrangement can also facilitate bonding with the substrate **52**, when applying an adhesive material on the flange part **23**.

(49) The thickness (**D2**) of the flange part **23** with respect to the thickness (**D1**) of the connection part **22** can be 1.1 times or greater, preferably 1.2 times or greater, more preferably 1.5 times or greater, and 3 times or less, preferably 2.5 times or less, more preferably 2 times or less.

(50) The thermosetting first resin used to form the cover part **2** is a light-transmissive thermosetting resin. Examples of the thermosetting resin include phenol resin, urea resin, melamine resin, epoxy resin, silicone resin, and polyurethane resin, and silicone resin is preferable. The use of a thermosetting resin that is more resistant to light and heat than a conventionally used thermoplastic resin such as polycarbonate resin can reduce degradation of the cover part **2**, and further can reduce degradation with the lapse of time, in which light and heat concentrate in portions discolored by degradation, where darkening of the cover part **2** progresses in an accelerative manner. Moreover, silicone resin exhibits high flowability when heated, which allows molding of the flange part **23** with a greater thickness.

Light-Shielding Part

(51) The light-shielding part **3** is applied to cover outer lateral surfaces **22a** of the lateral side walls

constituted by the connection part **22** and an upper surface **23a** of the flange part **23**, and is formed of a thermosetting second resin having a greater light-absorptance or a greater light-reflectance than the thermosetting first resin.

(52) The light-shielding part **3** is formed in a shape corresponding to the shapes of the connection part **22** and the flange part **23**, with a substantially uniform thickness. More specifically, the light-shielding part **3** has a shape in cross section corresponding to the shapes of the outer lateral surface **22a** of the lateral side walls constituted by the connection part **22** and the upper surface **23a** of the flange part **23**, which is, for example, an L-shape. The thickness of the light-shielding part **3** that is a thickness in a direction normal to the corresponding one of the lateral side walls constituted by the connection part **22** or normal to the flange part **23** is preferably in a range of 200 μm to 3,000 μm , more preferably in a range of 250 μm to 2,000 μm . With this arrangement, the outer lateral surfaces **22a** of the lateral side walls constituted by the connection part **22** and the upper surface **23a** of the flange part **23** can be reliably covered by the light-shielding part **3**, such that leaking of light through the outer lateral surfaces **22a** of the lateral side walls constituted by the connection part **22** and the upper surface **23a** of the flange part **23** can be substantially prevented. The light-shielding part **3** is disposed on the outer lateral surfaces **22a** of the lateral side walls constituted by the connection part **22** and on the upper surface **23a** of the flange part **23** without using an adhesive material. Because an adhesive material is not used, degradation or detachment of the adhesive material, and/or absorption of light by an adhesive material do not occur.

(53) The thermosetting second resin used to form the light-shielding part **3** is a thermosetting resin having a greater light-absorptance or a greater light-reflectance than the light-transmissive first resin. For the thermosetting second resin, a black colored resin or white colored resin of a light-transmissive thermosetting resin similar to that used as the thermosetting first resin, preferably a silicone resin, containing a black color material such as carbon having a high light absorptance, or a white color material such as titanium oxide having a high light reflectance can be used. With this, the connection part **22** formed of the light-transmissive first resin can be covered by the light-shielding part **3** formed of the thermosetting second resin having a high light absorptance or a high light reflectance, such that when the lens **1** is in operation, light from the light emitting element **51** is absorbed or reflected at the light-shielding part **3** and leaking of light in a lateral sides of the lens **1**, particularly, through the connection part **22** can be reduced.

Light Emitting Device

(54) Next, the light emitting device **100** will be described.

(55) The light emitting device **100** includes a light emitting element **51**, a lens **1**, and preferably a substrate **52**. A lens part **21** of the lens **1** is arranged at a position such that light from the light emitting element **51** can be transmitted through the lens part **21**. The light emitting device **100** may further include a light-transmissive member **53**. The light emitting device **100** includes the lens **1** whose outer lateral surfaces are covered by the light-shielding part **3**, such that light from the light emitting element **51** is absorbed or reflected by the light-shielding part **3**, and thus leaking of light in the lateral sides of the lens **1** can be reduced. The lens **1** is similar to that described above and therefore the description thereof will be appropriately omitted.

Light Emitting Element

(56) The light emitting element **51** preferably include at least a nitride-based semiconductor layered structure. The nitride-based semiconductor layered structure includes a first semiconductor layer (for example, an n-type semiconductor layer), a light emitting layer, and a second semiconductor layer (for example, a p-type semiconductor layer) layered in this order, which configured to generate light. The nitride-based semiconductor layered structure has a thickness of preferably 30 μm or less.

(57) The first semiconductor layer, the light-emitting layer, and the second semiconductor layer can be respectively an appropriate type, made of appropriate materials. Examples thereof include a Group compound semiconductor and a Group II-VI compound semiconductor. More specific

examples include nitride-based semiconductor materials such as $\text{In.sub.xAl}_{1-x}\text{Ga.sub.1-X-YN}$ ($0 \leq X$, $0 \leq Y$, $X+Y \leq 1$); for example, InN , AlN , GaN , InGaN , AlGaIn , or the like. For the thickness and structure of each of the layers, any appropriate thickness and structure known in the art can be employed. The light emitting element **51** generally has a quadrangular shape in a plan view, but can have any appropriate shape such as a circular shape, an elliptical shape, or a polygonal shape such as a triangular shape, a quadrangular shape, or a hexagonal shape.

Light-Transmissive Member

(58) It is preferable that an upper surface of the light emitting element **51** is covered by a light-transmissive member **53** disposed by using a spraying method or the like. The light-transmissive member **53** is configured to protect the light emitting element **51** from an external force, dust, moisture, or the like, and also to improve heat-resisting properties, weather resistant properties, and light-resisting properties of the light emitting element **51**. It is preferable that the sealing member can transmit 60% or greater light emitted from the light emitting layer. Such a light-transmissive member **53** can be formed of a thermosetting resin, a thermoplastic resin, a modified resin of such a resin, a hybrid resin which includes one or more of those resins, or the like. More specifically, an epoxy/modified epoxy resin, a silicone/modified silicone/hybrid silicone resin, or the like can be used.

(59) In order to adjust the color of emitted light, the light-transmissive member **53** preferably contains a fluorescent material to convert the wavelength of light from the light emitting element **51**. For the fluorescent material, a known material in the art can be used. Examples of the fluorescent material include yttrium aluminum garnet (YAG)-based fluorescent material activated with cerium.

(60) The light-transmissive member **53** may contain a filler material (for example, a diffusion agent, a coloring agent, or the like). Examples of the filler material include silica, titanium oxide, zirconium oxide, magnesium oxide, glass, a crystal or sintered body of a fluorescent material, and a sintered body of a fluorescent material and an inorganic binding material.

Substrate

(61) The substrate **52** is configured to mount a light emitting element **51**, and for example, includes a base material made of such as sapphire, spinel, or SiC , and a wiring pattern formed on or/and in the base material.

(62) The light emitting element **51** is preferably flip-chip mounted on the substrate **52**. On the substrate **52**, either a single or a plurality of light emitting elements **51** can be mounted. When a plurality of light emitting element **51** is mounted on the substrate **52**, the light emitting elements **51** may be arranged irregularly, or arranged regularly such as in rows and columns, or periodically. The type of connection used in connecting the plurality of light emitting elements **51** can be in series, in parallel, in series-parallel or in parallel-series.

Arrangement of Light Emitting Element and Lens

(63) The lens **1** is disposed such that the lens part **21** is at a position that allows light from the light emitting element **51** to be transmitted through the lens part **21**. The lens **1** is disposed on the base **52** through the flange part **23**, with the lens part **21** facing the light emitting element **51**. Accordingly, the lens **1** is disposed on the substrate **52** with the lens part **21** facing the light emitting element **51**, such that light from the light emitting element **51** leaking through the flange part **23** in lateral sides of the lens **1** can be reduced.

(64) The light emitting element **51** is arranged spaced apart from the lens part **21** and the connection part **22**. More specifically, a depth of the recess **4** defined by the lens part **21** and the lateral side walls constituted by the connection part **22** is in a range of about 0.4 mm to 1.5 mm. Accordingly, a gap between the bottom surface of the recess **4** and the upper surface of the light emitting element **51** is in a range of about 0.05 mm to 0.5 mm. With this arrangement, heat from the light emitting element **51** can be prevented from directly conducted to the lens part **21**, and thermal degradation of the lens part **21** can be reduced.

(65) In the lens **1**, the connection part **22** is formed such that a distance from the lens part (Fresnel lens) **21** to the light emitting element **51** is smaller than a distance from the connection part **22** to the light emitting element **51**. In the lens **1**, a recess **4** is defined by the lens part (Fresnel lens) **21** and the connection part **22** such that a distance from the lens part (Fresnel lens) **21** to the light emitting element **51** is smaller than a distance from the connection part **22** to the light emitting element **51**. More specifically, the center of the light emitting element **51** (or a geometrical center of the light emitting element **51**) is facing the center of the Fresnel lens **21** (or a geometrical center of the Fresnel lens **21**). In other words, the lens **1** is disposed such that the center of the ridges **21c** of the lens **1** arranged in concentric circles or concentric ellipsoids and the center of the light emitting element **51** are aligned. When the lens includes a plurality of lens parts **21**, that is, when the lens is a compound eye lens, the light emitting element **51** may be arranged with a shift such that the center of the light emitting element **51** is closer to the center of the entire lens. With the arrangements of the light emitting element **51** described above, a certain gap can be maintained between the light emitting element **51** and the connection part **22** and between the light emitting element **51** and the flange part **23**, such that light of the light emitting element **51** passing through the flange part **23** can be reduced.

(66) The lens **1** has the flange part **23** with a thickness (D2) greater than a thickness (D1) of the connection part **22** ($D1 < D2$), which allows reliable connection with an engaging part **54** (for example shown in FIG. 7A) provided at an outer side with respect to the designated disposing location of the lens **1**.

Method of Manufacturing Lens

(67) Next, a method of manufacturing a lens will be described. FIG. 5 is a flow chart showing a procedure of a method of manufacturing a lens according to the first embodiment. FIG. 6A is a cross-sectional view schematically illustrating forming a cover blank in a method of manufacturing a lens according to the first embodiment. FIG. 6B is a cross-sectional view schematically showing removing a first mold in a method of manufacturing a lens according to the first embodiment. FIG. 6C is a cross-sectional view schematically illustrating arranging a cover blank in a second mold and forming a lens blank in a method of manufacturing a lens according to the first embodiment. FIG. 6D is a cross-sectional view schematically illustrating cutting a lens blank to obtain individual lenses in a method of manufacturing a lens according to the first embodiment.

(68) Although FIG. 6A to FIG. 6D illustrate manufacturing of a plurality of lenses, manufacturing of a single lens will be described below, and corresponding description of manufacturing adjacent lenses will be appropriately omitted.

(69) The method of manufacturing a lens **1** includes forming (S1) a cover blank, removing (S2) a first mold, arranging (S3) the cover blank in a second mold, forming (S4) a lens blank, and obtaining (S5) individual lenses, which are performed in this order. Accordingly, the lens **1** allowing for a reduction of leakage of light can be obtained. Forming the cover blank **20** with a thickness (D1) of the connection part **22** smaller than a thickness (D2) of the flange part **23**, the lens **1** can be stably connected to an outer engaging part **54** (shown in FIG. 7A).

(70) The description of material and arrangement etc., of each member are as in the lens **1** and therefore the description thereof will be appropriately omitted.

Forming Cover Blank

(71) In forming (S1) a cover blank, a thermosetting first resin is injected in a first mold and cured to form a cover blank **20** having a plurality of cover parts **2** each having a lens part **21** having lateral sides, a connection part **22** constituting lateral side walls each extending downward from a respective one of the lateral sides of the lens part **21**, and a flange part **23** extending outward from a lower-end portion of the lateral side walls constituted by the connection part **22**. The lens part **21**, the connection part **22**, and the flange part **23** are formed continuous with one another.

(72) In the forming (S1) a cover blank, an upper mold **61** and a lower mold **62** adapted for transfer molding are used as the first mold. The upper mold **61** and the lower mold **62** are closed, the

thermosetting first resin, which is heated and softened, is injected under pressure into the first mold through a resin injecting port **61a** formed in the upper mold **61**. The thermosetting first resin is cured in the heated mold, such that a cover blank **20** having a plurality of cover parts **2** connected one another at the flange part **23** is molded.

(73) In the forming (S1) a cover blank, the upper mold **61** and the lower mold **62** are designed to produce a cover blank **20** having the plurality of cover parts **2** each formed with a recess having an opening facing downward, which is defined by the lens part **21** and the connection part **22** and is located inward of the flange part **23**. Further, in the forming (S1) a cover blank, the upper mold **61** and the lower mold **62** are designed to produce a cover blank **20** having a plurality of cover parts **2**, a respective one of which has a lens part **21** having a plurality of ridges **21c** on both sides, with the flange part **23** having a thickness (D2) greater than a thickness (D1) of the connection part **22**, for example, a thickness of the flange part **23** greater than 30 μm .

Removing First Mold

(74) In the removing (S2) the first mold, a part or all parts of the first mold is removed after the cover blank **20** is formed. In the first embodiment, the upper mold **61** that is a part of the first mold is removed and the cover blank **20** is held by the lower mold **62**.

Arranging Cover Blank in Second Mold

(75) In the arranging (S3) the cover blank in the second mold, another upper mold **71** which is also adapted for transfer molding is arranged on the lower mold **62** holding the cover blank **20**. In the arranging (S3) the cover blank in the second mold, the upper mold **71** and the lower mold **62** are used as the second mold.

Forming Lens Blank

(76) In the forming (S4) a lens blank, a thermosetting second resin having a greater light-absorptance or a greater light-reflectance than the thermosetting first resin is injected in the second mold and cured, to form a lens blank **10** having a light-shielding part **3** between adjacent cover parts **2**.

(77) In the forming (S4) a lens blank, the upper mold **71** and the lower mold **62** are closed, and the thermosetting second resin, which is heated and softened, is injected under pressure into the second mold through a resin injecting port **71a** formed in the upper mold **71**. The thermosetting second resin is cured in the heated mold, such that a lens blank **10** having a light-shielding part **3** between adjacent cover parts **2**, in a bent shape in a cross-section that is in conformity to the shapes of the outer lateral surfaces **22a** of the lateral side walls constituted by the connection part **22** and the upper surface **23a** of the flange part **23** is molded.

Obtaining Individual Lenses

(78) As shown in FIG. 6D, in the obtaining (S5) individual lenses, the lens blank **10** is singulated into individual lenses **1**. That is, in the obtaining (S5) individual lenses, all parts of the second mold are removed to take out the lens blank **10**, and the lens blank **10** is cut along a center of the light-shielding part **3** between adjacent cover parts **2**. The singulating is performed by cutting at the light-shielding part **3**, thus obtaining individual lenses **1** each having the light-shielding part **3** covering the outer lateral surfaces of the lateral side walls constituted by the connection part **22** and the upper surface **23a** of the flange part **23**.

(79) Further, in the obtaining (S5) individual lenses, a center of the light-shielding part **3** between outer lateral surfaces **22a** of the lateral side walls constituted by two adjacent connection parts **22** of the lens blank **10** is cut together with the flange part **23** to obtain an individual lens **1** having the outer lateral surfaces **22a** of the lateral side walls constituted by the connection part **22** and the upper surface **23a** of each of the one or more flange parts **23** covered by the light-shielding part **3**. In a cross-section, the light-shielding part **3** of each of the individual lenses **1** has a rectangular shape in conformity to the shapes of the outer lateral surfaces **22a** of the lateral side walls constituted by the connection part **22** and the upper surface **23a** of the flange part **23**. Cutting of the light-shielding part **3** can be performed by using a cutting tool **91** such as a blade or the like known

in the art. It is preferable to adjust the cutting width of the cutting tool **91** such that the flange part **23** of the lens **1** has a length in a range of 200 μm to 1,000 μm .

Method of Manufacturing Light Emitting Device

(80) Next, a method of producing a light emitting device **100** will be described. Although the method of manufacturing the light emitting device **100** is not shown in the drawings, the method will be described with reference to FIG. **1** and FIG. **4** that show the structure of the light emitting device **100**.

(81) The method of manufacturing the light emitting device **100** includes providing a lens and disposing the lens with respect to the light emitting element that has been positioned.

(82) The providing a lens is similar to that described in the method of manufacturing a lens and therefore the description thereof will be appropriately omitted.

Disposing Lens

(83) In the disposing a lens, the lens **1** is disposed such that light from the light emitting element **51** to be transmitted through the lens part **21** of the lens **1**. In the disposing a lens, the lens **1** is disposed such that the light emitting element **51**, which is preferably has been mounted on a substrate **52**, is accommodated in the recess **4** having an opening facing downward. The recess **4** is defined by the lens part **21** and the connection part **22**. The lens part **21** defines a bottom surface of the recess **4** and the connection part **22** defines lateral surface(s) of the recess **4**. The lateral surface(s) define the opening of the recess **4**. The opening of the recess **4** is located inward of the flange part **23**. The light emitting element **51** may be covered by a light-transmissive member **53**.

(84) In the disposing a lens, it is preferable that the light emitting element **51** is mounted on a substrate **52**, in which the light emitting element **51** is flip-chip mounted on the substrate **52** by using soldering. It is preferable that after the light emitting element **51** is mounted on the substrate **52**, the flange part **23** is connected to the substrate **52**. In the disposing a lens, the lens **1** is disposed such that the connection part **22** and the lens part **21** of the lens **1** are spaced apart from the light emitting element **51**. More specifically, a gap between the bottom surface of the recess **4** and the upper surface of the light-transmissive member **51** or the upper surface of the light-transmissive member **53** is preferably in a range of 0.05 mm to 0.5 mm. Further, in the disposing a lens, the connection part **22** (recess **4**) is preferably formed such that a distance from the lens part **21** to the light emitting element **51** is smaller than a distance from the connection part **22** to the light emitting element **51**. More specifically, the center of the light emitting element **51** (or a geometrical center of the light emitting element **51**) is preferably aligned with the center of the lens part (the Fresnel lens) **21** (or a geometrical center of the lens part (Fresnel lens) **21**), that is, aligned with the centers of the ridges **21c** arranged in concentric circles or concentric ellipsoids.

(85) When the lens **1** is disposed such that the light emitting element **51** is accommodated in the recess **4**, the lens **1** may be disposed via an engaging part **54** as shown in FIG. **7A** to FIG. **7C**. FIG. **7A** is a schematic cross-sectional diagram illustrating a state of arranging a lens according to the first embodiment via an engaging member with respect to a substrate having a light emitting element mounted thereon. FIG. **7B** is a cross-sectional diagram illustrating a state of engaging a flange part of a lens according to the first embodiment with an engaging member. FIG. **7C** is a schematic cross-sectional view illustrating a lens arranged on a substrate, in which a flange part of a lens according to the first embodiment is engaged with an engaging part. The substrate **52** may be provided beforehand with the engaging part **54** at a predetermined location around the light emitting element **51**. The engaging part **54** has an inwardly projecting end portion and a shaft portion, and the end portion has a thickness greater than that of a shaft portion. The engaging part **54** is disposed such that a tip of the inwardly projecting end portion is at a location corresponding to substantially a half of protruding length of the flange part **23**. Thus, together with the light-shielding part **3**, the flange part **23** can be engaged with the engaging part **54**. Either the engaging part **54** or the flange part **23** of the lens **1** is elastically deformed (in FIG. **7B**, the engaging part **54** is elastically deformed) to engage the flange part **23** (the flange part **23** and the light-shielding part

3) with the engaging part **54**. Thus, the lens **1** is disposed on the substrate **52**. In the lens **1**, the flange part **23** has a thickness greater than a thickness of the connection part **22**, which allows stable attachment to the engaging part **54**.

Second Embodiment

(86) Next, a second embodiment will be described with reference to FIG. **8A**.

(87) FIG. **8A** is a cross-sectional view schematically showing a structure of a light emitting device that includes a lens according to a second embodiment, taken along a line passing through the center of the light emitting device. The same reference numerals may be applied to the components that have been described above and description thereof may be appropriately omitted.

Lens and Light Emitting Device

(88) A lens **1A** and a light emitting device **100A** will be described.

(89) In the lens **1A** and the light emitting device **100A** according to the second embodiment, a light-shielding part **3A** has a large thickness and a rectangular cross-sectional shape which is in conformity to outer lateral surfaces **22a** of the lateral side walls constituted by the connection part **22**. This is different from the connection part **22** of the first embodiment, which has a cross-sectional shape bent along the shapes of the outer lateral surfaces **22a** of the lateral side walls constituted by the connection part **22** and the shape of the upper surface **23a** of the flange part **23**. Other structures are similar to the lens **1** and the light emitting device **100** according to the first embodiment. The light-shielding part **3A** is disposed with the same thickness as the length of the flange part **23** extending outward from the connection part **22**. The outer lateral surface of the light-shielding part **3A** and the outer end surface of the flange part **23** are flush with each other, which can reduce occurrence of detachment at the interface between the light-shielding part **3A** and the flange part **23**. The light-shielding part **3A** preferably covers an entire outer lateral surfaces **22a** of the lateral side walls constituted by the connection part **22**.

(90) The lens **1A** and the light emitting device **100A** according to the second embodiment have the light-shielding part **3A** with a larger thickness as described above, and thus leaking of light can be further reduced. Also, the flange part **23** has a larger thickness than that of the connection part **22**, which together with the light-shielding part **3A**, can enhance mechanical strength, such that when the cover part **2** is bonded to the substrate **52** via an adhesive material, stable bonding can be obtained.

Obtaining Individual Lenses

(91) Next, a method of obtaining individual lenses **1A** will be described.

(92) FIG. **8A** is a cross-sectional view schematically illustrating forming a cover blank in a method of manufacturing a lens according to the second embodiment. FIG. **8C** is a cross-sectional view schematically illustrating cutting a lens blank into individual lenses in a method of manufacturing a lens according to the second embodiment.

(93) The obtaining individual lenses **1A** includes, similar to that in the obtaining individual lenses according to the first embodiment, forming (S1) a cover blank, removing (S2) a first mold, arranging (S3) the cover blank in a second mold, forming (S4) a lens blank, and obtaining (S5) individual lenses, which are performed in this order. The first mold and the second mold are designed such that the flange part **23** has a thickness greater than that of the connection part **22**.

(94) The obtaining individual lenses **1A** is similar to the obtaining individual lenses **1** according to the first embodiment except that, in the arranging (S3) a cover blank **20** in the second mold, the cover blank **20** is arranged in a second mold (an upper mold **81** and a lower mold **62**) that is different from the second mold (the upper mold **71** and the lower mold **62**) used in the first embodiment. In the forming (S4) a lens blank, a thermosetting second resin having a greater light-absorptance or a greater light-reflectance than the thermosetting first resin is injected in the second mold and cured, to form a lens blank **10A** having a light-shielding part **3A** having a rectangular cross-sectional shape that is in conformity to the outer lateral surfaces **22a** of the lateral side walls constituted by the connection part **22**, between adjacent cover parts **2**. The thermosetting second

resin is injected from a resin injecting port **81Aa** of the upper mold **81A**. Further, in the obtaining individual lenses **1A**, a center of the light-shielding part **3A** between outer lateral surfaces **22a** of the lateral side walls constituted by two adjacent connection parts **22** of the lens blank **10A** is cut together with the flange part **23** to obtain individual lenses **1A** each having the outer lateral surfaces **22a** of the lateral side walls constituted by the connection part **22** and the upper surface **23a** of the flange part **23** covered by the light-shielding part **3**. In a cross-section, the light-shielding part **3A** of the individual lens **1A** thus obtained has a rectangular shape in conformity to the shape of the outer lateral surface **22a** of the connection part **22**.

Method of Manufacturing Light Emitting Device

(95) Next, a method of manufacturing a light emitting device **100B** will be described. The method of manufacturing the light emitting device **100B** includes providing a lens **1A** and disposing the lens **1A** which are performed in this order. The method of manufacturing the light emitting device **100B** can be performed in a similar manner as in the method of manufacturing the light emitting device **100** according to the first embodiment, except for using the lens **1B**.

Third Embodiment

(96) Next, a third embodiment will be described with reference to FIG. **9**.

(97) FIG. **9** is a cross-sectional view schematically showing a structure of a light emitting device that includes a lens according to a third embodiment, taken along a line passing through the center of the light emitting device. In the description below, the same reference numerals will be applied to the structure of the flexible substrate member **1** and description thereof will be omitted.

Lens and Light Emitting Device

(98) A lens **1B** and a light emitting device **100B** will be described. A flange part **23** has a thickness greater than a thickness of a connection part **22**.

(99) In the lens **1B** and the light emitting device **100B** according to the third embodiment, a light-shielding part **3B** has a thickness greater than the thickness of the light-shielding part **3A** of the second embodiment, and the shielding part **3B** also covers an outer end surface **23b** of the flange part **23**. Others are similar to those of the method of manufacturing the lens **1A** and the light emitting device **100A**, according to the second embodiment.

(100) The lens **1B** and the light emitting device **100B** according to the third embodiment have the light-shielding part **3B** having a greater thickness that covers the outer lateral end surface **23b** of the flange part **23**, thus yielding a further reduction of leaking light through the light-shielding part **3B**.

Method of Manufacturing Lens

(101) Next, a method of manufacturing a light Lens **100B** will be described.

(102) FIG. **10** is a flow chart showing a procedure of a method of manufacturing a lens according to the first embodiment. FIG. **11A** is a cross-sectional view schematically illustrating forming a cover blank in a method of manufacturing a lens according to the third embodiment. FIG. **11B** is a cross-sectional view schematically illustrating cutting a cover blank in a method of manufacturing a lens according to the third embodiment. FIG. **11C** is a cross-sectional view schematically illustrating arranging a cut cover blank in a second mold and forming a lens blank in a method of manufacturing a lens according to the third embodiment. FIG. **11D** is a cross-sectional view schematically illustrating cutting a lens blank to obtain individual lenses in a method of manufacturing a lens according to the third embodiment.

(103) The method of manufacturing a lens **1B** includes forming (S**11**) a cover blank, removing (S**12**) a first mold, cutting (S**13**) the cover blank, arranging (S**14**) the cover blank in a second mold, forming (S**15**) a lens blank, and obtaining (S**16**) individual lenses, which are performed in this order. With the method of manufacturing described above, a lens **1B** yielding a reduction of leaking light can be produced.

(104) The materials and arrangements of the members are similar to those described for the lens **1B** and therefore the description thereof will be appropriately omitted.

Forming Cover Blank

(105) As shown in FIG. 11A, in the forming (S11) a cover blank, a thermosetting first resin is injected in a first mold and cured to form a cover blank **20** having a plurality of cover parts **2** each having a lens part **21** having lateral sides, connection part **22** constituting lateral side walls each extending downward from a respective one of the lateral sides of the lens part **21**, and a flange part **23** extending outward from a lower-end portion of the lateral side walls constituted by the connection part **22**. The lens part **21**, the connection part **22**, and the flange part **23** are formed continuous with one another.

(106) In the forming (S11) a cover blank, an upper mold **61** and a lower mold **62** adapted for transfer molding are employed as the first mold. The upper mold **81A** and the lower mold **82A** are closed, and the thermosetting second resin, which is heated and softened, is injected under pressure into the second mold through a resin injecting port **61a** provided in the upper mold **61**. The thermosetting first resin is cured in the heated mold, such that a cover blank **20** having a plurality of cover parts **2** connected to one another at the flange parts **23** is molded.

(107) In the forming (S11) a cover blank, the upper mold **61** and the lower mold **62** are designed to produce a cover blank **20** having a plurality of cover parts **2** each formed with a recess having an opening facing downward. The recess is defined by the lens part **21** and the connection part **22** with an opening located inward of the flange part **23**. Each of the lens parts **21** of the cover blank **20** has a plurality of ridges **21c** at both upper and lower sides. Further, in the forming (S11) a cover blank, the upper mold **61** and the lower mold **62** are designed to produce a cover blank **20** having a plurality of cover parts **2** each having the flange part **23** with a thickness greater than 30 μm , which is greater than a thickness of the connection part **22**.

Removing First Mold

(108) In the removing (S12) the first mold, the first mold is removed after the cover blank **20** is formed. In this step, all parts (the upper mold **61** and the lower mold **62**) of the first mold are removed.

Cutting Cover Blank

(109) As shown in FIG. 11B, in the cutting (S13) the cover blank, the cover blank **20** is cut at the flange part **23** between adjacent cover parts **2** to obtain individual cover blanks **20A**. Cutting of the cover blank **20** can be performed by using a cutting tool **91** such as a known blade or the like. It is preferable to adjust the cutting width of the cutting tool **91** such that the flange part **23** of the individual cover blank **20A** has a length in a range of 200 μm to 1000 μm .

Arranging Cover Blank in Second Mold

(110) In the arranging (S14) the cover blank in the second mold, cut cover blank **20** (singulated cover blank **20A**) is arranged in another set of mold (an upper mold **81A** and a lower mold **82A**) adapted for transfer molding.

Forming Lens Blank

(111) In the forming (S15) a lens blank, a thermosetting second resin having a greater light-absorptance or a greater light-reflectance than the thermosetting first resin is injected in the second mold and cured, to form a lens blank **10B** having a light-shielding part **3** between adjacent cover parts **2**. In the forming (S15) a lens blank, the upper mold **81A** and the lower mold **82A** are closed, and the thermosetting second resin, which is heated and softened, is injected under pressure into the second mold through a resin injecting port **81Aa** formed in the upper mold **81A**. The thermosetting second resin is cured in the heated mold, such that a lens blank **10B** having a light-shielding part **3B** between adjacent cover parts **2**, in a rectangular shape in a cross-section that is in conformity to the shape of the outer lateral surface **22a** of the side walls constituted by the connection part **22** and covering an upper surface **23a** and an outer end surface **23b** of the flange part **23** is molded.

Obtaining Individual Lenses

(112) As shown in FIG. 11D, in the obtaining (S16) individual lenses, the lens blank **10B** is singulated into individual lenses **1B**. That is, in the obtaining (S16) individual lenses, all parts of

the second mold are removed to take out the lens blank **10B**, and the lens blank **10B** is cut along a center of the light-shielding part **3B** between adjacent cover parts **2**. The singulating is performed by cutting at the light-shielding part **3B**, thus obtaining individual lenses **1B** each having the light-shielding part **3B** covering the outer lateral surfaces **22a** of the lateral side walls constituted by the connection part **22** and also covering the outer end surfaces **23b** of the flange part **23**.

(113) In the obtaining (**S16**) individual lenses, the lens blank **10B** is cut along the center of the light-shielding part **3B** between the outer lateral surfaces **22a** of the lateral side walls constituted by the connection part **22** and between outer end surfaces **23b** of adjacent flange parts **23** of the lens blank, to obtain individual lenses **1B** each having the light-shielding part **3** covering the outer lateral surfaces **22a** of the lateral side walls constituted by the connection part **22**, the upper surface **23a** of the flange part **23**, and the outer end surfaces of the flange part **23**. In a cross-section, the light-shielding part **3B** of each of the individual lenses **1B** has a rectangular shape in conformity to the shapes of the outer lateral surfaces **22a** of the lateral side walls constituted by the connection part **22** and covering the outer end surface **23b** of the flange part **23**. Cutting of the light-shielding part **3B** can be performed by using a cutting tool **91** such as a known blade or the like.

Method of Manufacturing Light Emitting Device

(114) Next, a method of manufacturing a light emitting device **100B** will be described.

(115) The method of manufacturing the light emitting device **100B** includes providing a lens and disposing the lens, which can be performed in this order. The method of manufacturing the light emitting device **100B** can be performed in a similar manner as in the method of manufacturing the light emitting device **100** according to the first embodiment, except for using the lens **1B**.

Fourth Embodiment, Fifth Embodiment

(116) In the first embodiment to the third embodiment, structures including the flange part **23** provided along a lower end-portion (e.g., an opening-end portion) of entire outer periphery of the lateral side walls constituted by the connection part **22** have been illustrated, but as shown in FIG. **12A**, FIG. **12B**, FIG. **13A**, and FIG. **13B**, one or more flange parts **23d** may be formed such that each flange part **23d** is formed extending outward from a lower end portion of a corresponding one of the lateral side walls constituted by the connection part **22**. FIG. **12A** is a plan view schematically showing a structure of a lens according to a fourth embodiment. FIG. **12B** is a schematic cross-sectional view taken along line XIIB-VIIB of FIG. **12A**. FIG. **13A** is a plan view schematically showing a structure of a lens according to a fifth embodiment. FIG. **13B** is a schematic cross-sectional view taken along line XIIIB-XIIIB of FIG. **13A**.

(117) As shown in FIG. **12A** and FIG. **12B**, in the lens **1D** and the light emitting device **100D** according to the fourth embodiment, each flange part **23d** is formed extending outward from a lower end portion of a corresponding one of the outer lateral surfaces of the lateral side walls of the connection part **22** along a partial length thereof. With this structure, the volume of the flange part **23d** can be reduced. That is, the surface area of the flange part **23d** can be reduced. Further, the lens **1D** includes a light-shielding part **3d** covering an upper surface and lateral end surfaces of each flange part **23d** and the lateral outer surfaces of the lateral side walls constituted by the connection part **22**. Accordingly, when the lens **1D** is in operation, leaking of light through lateral sides of the lens **1D**, particularly leaking of light through the flange parts **23d** can be further reduced.

(118) The flange parts **23d** are formed at locations such that, in a plan view, each two opposite flange parts **23d** are substantially symmetric about a plane parallel to their corresponding ones of the lateral side walls constituted by the connection part **22** and is passing through the center of the lens part **21**. The flange parts **23d** have a thickness greater than that of the connection part **22**. The flange parts **23d** are arranged one on the lower end portion of each of the outer lateral surfaces of the lateral side walls of the connection part **22**, such that four flange parts **23d** are arranged on the outer periphery of the connection part **22** each along a partial length thereof. Each of the flange parts **23d** is located substantially central to the width of the outer lateral surface of the corresponding one of the side walls constituted by the connecting part **22**. The flange parts **23d**

each preferably have a width in a range of 50 μm to 1,000 μm . The flange parts **23d** each have a thickness of preferably 30 μm or greater, more preferably 50 μm , or 100 μm .

(119) With the structure as described above, more stable bonding or fixing to the substrate **52** can be achieved in the lens **1D** and the light emitting device **100D** according to the fourth embodiment.

(120) As shown in FIG. **13A** and FIG. **13B**, in a lens **1E** and a light emitting device **100E** according to a fifth embodiment, two flange parts **23e** are formed on a lower end portion of each of the outer lateral surfaces of the lateral side walls constituted by the connection part **22**, such that eight flange parts **23e** are arranged on the outer periphery of the connection part **22** each along a partial length thereof. The flange parts **23e** formed on the outer lateral surface of a single lateral side wall are spaced apart from each other. Each two spaced apart flange parts **23e** formed on one outer lateral surface of the connection part **22** and the two spaced apart flange parts **23e** formed on the outer lateral surface opposite the one outer lateral surface are respectively located opposite from each other. Other structures are similar to that in the lens **1** and the light emitting device **100** according to the first embodiment.

(121) In the fifth embodiment, two flange parts **23e** are disposed on each of the outer lateral surfaces of the lateral side walls constituted by the connection part **22**, but three or four flange parts **23e** may be formed on each of the outer lateral surfaces of the lateral side walls constituted by the connection part **22**. Different pair of opposite outer lateral surfaces of the connection part **22** may be provided with a different number of the flange parts **23e**.

Sixth Embodiment

(122) For example, as the lens **1F** and the light emitting device **100F** shown in FIG. **14**, a lens part **21f** may be a compound eye lens having a plurality of Fresnel lenses. FIG. **14** is a plan view schematically showing a structure of a lens **1F** according to a sixth embodiment. The lens **1F** has a lens part **21f** including a circular lens functioning part at a center of a rectangular shape in a plan view, and a flat plate-like shape around surrounding the lens functioning parts. The circular lens part in the lens part **21f** is cut in four, and a respective one of a quarter of the circular part is formed into a Fresnel lens. It is preferable that light emitting elements **51** of a number corresponding to the number of the Fresnel lens are disposed such that a center of each of the light emitting elements **51** is offset toward the center of the lens part **21f** from the center of corresponding Fresnel lens of the lens part **21f**. When the lens part **21f** has a compound eye lens, disposing the light emitting elements **51** with the centers offset from the centers of the respective Fresnel lenses as described above allows for an increase of the light extraction efficiency. It is further preferable that two types of the light emitting elements **51** to emit two different emission colors are employed and the light emitting elements **51** of the same emission color are disposed diagonally. The lens **1F** has a structure similar to that shown in FIG. **1**, except that the flange part **23f** has a thickness greater than that of the connection part **22**. The flange part **23f** may be provided partially along the outer lateral surfaces of the side walls constituted by the connection part **22** as in the other embodiments.

Seventh Embodiment

(123) As shown in FIG. **15**, a plano-convex Fresnel lens having a plurality of ridges **21c** on either upward-facing surface or downward-facing surface (in FIG. **15**, the light-incident surface **21a**) may be used in the lens part **21g** of the lens **1G** and the light emitting device **100G**. FIG. **15** is a cross-sectional view schematically showing a structure of the light emitting device according to a seventh embodiment. When a compound eye lens is employed, a structure having the plurality of projections **21c** either in the upward-facing surface or the downward-facing surface of the lens part **21g** can also be employed. That is, it is preferable that the lens part **21g** has a light incident surface **21a** with a plurality of ridges **21c**, and a flat or substantially flat light-emitting surface **21b** such that the light-emitting surface **21b** of the lens part **21g** can be arranged in parallel to the light emitting surface of the light emitting element **51**. The light-emitting surface **21b** may have microscopic projections formed by emboss processing, mat processing, or the like, which are lower than the ridges **21c**. With this, light emitted through the light-emitting surface **21b** can be uniformly

dispersed. As in the lens **1G** and the light emitting device **100G**, employing a structure having a plurality of ridges **21c** only on the light-incident surface **21a**, allows for a reduction in a dimension in height direction, because the upward-facing surface does not have the ridges **21c**.

Eighth Embodiment

(124) As shown in FIG. **16**, a lens **1H** and a light emitting device **100H** may have a structure in which a connection part **22h** is located only along on a lateral side of the lens part **21** and a flange part **23h** is formed extending downward from the connection part **22** with a thickness greater than the thickness of the connection part **22h**. FIG. **16** is a cross-sectional view schematically showing a structure of the light emitting device according to an eighth embodiment.

(125) In the lens **1H**, outer lateral surfaces of the connection part **22** and upper surfaces and outer lateral sides of the flange part **23h** are covered by a light-shielding part **3h**. In the lens **1H**, the flange part **23h** and the lens part **21h** define a recess **4h**. In the lens **1H**, the lower surface of the flange part **23** and the lower surface of the light-shielding part **3h** are bonded to the substrate **52** through an adhesive such that the lens part **21h** faces the light emitting element **51** mounted on the substrate **52**. In the lens **1H**, the flange part **23h** has a thickness greater than that of the connection part **22h**, for example, the flange part **23h** has a thickness in a range of 30 μm to 1,000 μm .

(126) In the light emitting device **100H**, the light emitting element **51** mounted on the substrate **52** is spaced apart from the lens part **21h** of the lens **1H** depending on the height of the flange part **23h** of the lens **1H**.

Variant Example

(127) As shown in FIG. **17A** and FIG. **17B**, the lens and the light emitting device may include the flange part at different position(s) or in different numbers and/or shape(s) than those described above. Alternatively, as shown in FIG. **17C**, the lens part and the connection part may have a shape other than a quadrangular shape in a plan view. FIG. **17A** is a schematic plan diagram illustrating a variant example of a flange part of a lens and illustrating an engaged state of the flange part with an engaging part. FIG. **17B** is a schematic plan diagram illustrating another variant example of a flange part of a lens and illustrating an engaged state of the flange part with an engaging part. FIG. **17C** is a schematic plan diagram illustrating a variant example of a lens and illustrating an engaged state of the lens with an engaging part.

(128) As shown in FIG. **17A**, the lens **101A** may have four flange parts **123a** on four sides of the connection part respectively, with a width equivalent to the width of the connection part. Alternatively, as shown in FIG. **17B**, the lens **101A** may have two flange parts **123b** on opposite two sides of the connection part respectively, with a width smaller than the width of the connection part. The flange parts **123a** and the flange parts **123b** may be respectively configured for slide engagement with engaging parts **154A** and engaging parts **154B** provided on the substrate, in the respective light emitting devices.

(129) As shown in FIG. **17C**, the lens **101C** can be formed with the lens part **121c** in a circular shape in a plan view, the connection part **122c** in a cylindrical shape, and four flange parts **123c** provided every 90 degrees with respect to the outer periphery of the connection part **122c** in a plan view, such that respective two flange parts **123c** are opposite to each other. The outer lateral sides of the connection part **122c** and the upper surface and lateral end surfaces of each of the flange parts **123c** are covered by the light-shielding part **103c**. As illustrated above, the lens part **121c** can be formed in a shape other than a quadrangular shape and the connection part **122c** can be disposed in conformity with the shape of the lens part **121c**, at the outer edge of the lens part **121c** and downward. The light emitting device can be assembled by rotatively sliding the lens **101C** such that the flange parts **123c** are engaged with respective engaging parts **154C** provided on the substrate. In the variant example illustrated in FIG. **17C**, the lens part **121c** has a circular shape, but the lens part **121c** can be in a triangular shape or a pentagonal or higher polygonal shape. In each variational example, the location of the light-shielding part may be, as has been described above. The lens and light emitting device of each variational example can be manufactured according to the methods

described above, with appropriately changing respective molds.

(130) The lens and light emitting device, and the method of manufacturing the lens and the method of manufacturing the light emitting device described above are not limited to those embodiments described above.

(131) It is to be understood that although the present invention has been described with regard to preferred embodiments thereof, various other embodiments and variants may occur to those skilled in the art, which are within the scope and spirit of the invention, and such other embodiments and variants are intended to be covered by the following claims.

Claims

1. A method of manufacturing a lens, the method comprising: forming a cover blank, the forming a cover blank comprising, injecting a thermosetting first resin in a first mold and curing the thermosetting first resin, to form a cover blank having a plurality of cover parts, each of the cover parts including a lens part having one or more lateral sides, a connection part constituting one or more lateral side walls each extending downward from a respective one of the one or more lateral sides of the lens part, and one or more flange parts each extending outward from a lower-end portion of a corresponding one of the one or more lateral side walls, the lens part, the flange part, and the connection part being continuous to one another, the lens part and the connection part defining a recess having an opening facing downward, the one or more flange parts extending outward from a periphery of the opening of the recess; removing a part or all parts of the first mold; arranging the cover blank in a second mold; forming a lens blank, the forming a lens blank comprising, injecting a thermosetting second resin having a greater light absorptance or a greater light reflectance than the thermosetting first resin into the second mold, and curing the thermosetting second resin, to form a lens blank having a light-shielding part between adjacent ones of the cover parts; and obtaining individual lenses, the obtaining individual lenses comprising, taking out the lens blank from the second mold, and cutting the lens blank at the light-shielding part located between adjacent ones of the cover parts to obtain individual lenses each with an entirety of outer surfaces of the one or more lateral side walls covered by the light-shielding part and each with an entirety of upper surfaces of the one or more flange parts covered by the light-shielding part.

2. A method of manufacturing a lens, the method comprising: forming a cover blank, the forming a cover blank comprising, injecting a thermosetting first resin in a first mold, and curing the thermosetting first resin, to form a cover blank having a plurality of cover parts, each of the cover parts including a lens part having one or more lateral sides, a connection part constituting one or more lateral side walls each extending downward from a respective one of the one or more lateral sides of the lens part, and one or more flange parts each extending outward from a lower-end portion of a corresponding one of the one or more lateral side walls, the lens part, the one or more flange parts, and the connection part being continuous to one another, the one or more flange parts having a thickness greater than 30 μm ; removing a part or all parts of the first mold; arranging the cover blank in a second mold; forming a lens blank, the forming a lens blank comprising, injecting a thermosetting second resin having a greater light absorptance or a greater light reflectance than the thermosetting first resin into the second mold and curing the thermosetting second resin, to form a lens blank having a light-shielding part between adjacent ones of the cover parts; and obtaining individual lenses, the obtaining individual lenses comprising, taking out the lens blank from the second mold, and cutting the lens blank at the light-shielding part located between the adjacent ones of the cover parts to obtain individual lenses each with an entirety of outer surfaces of the one or more lateral side walls covered by the light-shielding part and each with an entirety of upper surfaces of the one or more flange parts covered by the light-shielding part.

3. A method of manufacturing a lens, the method comprising: forming a cover blank, the forming a

cover blank comprising, injecting a thermosetting first resin in a first mold, and curing the thermosetting first resin, to form a cover blank having a plurality of cover parts, each of the cover parts including a lens part having one or more lateral sides, a connection part constituting one or more lateral side walls each extending downward from a respective one of the one or more lateral sides of the lens part, and one or more flange parts each extending from a lower-end portion of a corresponding one of the one or more lateral side walls, each of the one or more flange parts having a greater thickness than the connection part, and the lens part, the one or more flange parts, and the connection part being continuous to one another; removing the first mold; cutting the cover blank at the flange part located between adjacent ones of the cover parts; arranging the cut cover blank in a second mold; forming a lens blank, the forming the lens blank comprising, injecting a thermosetting second resin having a greater light absorptance or a greater light reflectance than the thermosetting first resin into the second mold and curing the thermosetting second resin to form a lens blank having a light-shielding part located between adjacent ones of the cover parts; and obtaining individual lenses, the obtaining individual lenses comprising, taking out the lens blank from the second mold, cutting the lens blank at the light-shielding part located between the adjacent ones of the cover parts to obtain individual lenses each with an entirety of outer surfaces of the one or more side walls covered by the light-shielding part, each with an entirety of upper surfaces of the one or more flange parts covered by the light-shielding part, and each with an outer end surface of each of the one or more flange parts covered by the light-shielding part.

4. The method of manufacturing a lens according to claim 2, wherein in the forming a cover blank, the lens part, the connection part, and the one or more flange parts of each cover part are formed such that a recess having an opening is defined by the lens part and the connection part and the one or more flange parts extends from a periphery of the opening of the recess.

5. The method of manufacturing a lens according to claim 3, wherein in the forming a cover blank, the lens part, the connection part, and the one or more flange parts of each cover part are formed such that a recess having an opening is defined by the lens part and the connection part and the one or more flange parts extends from a periphery of the opening of the recess.

6. The method of manufacturing a lens according to claim 1, wherein in the forming a cover blank, the thermosetting first resin is a silicone resin.

7. The method of manufacturing a lens according to claim 1, wherein in the forming a lens blank, the thermosetting second resin is a silicone resin.

8. The method of manufacturing a lens according to claim 1, wherein in the forming a lens blank, the light-shielding part is disposed directly on the adjacent ones of the cover parts without using an adhesive material.

9. A method of manufacturing a light emitting device, the method comprising: providing a lens using a method according to claim 1, and disposing the lens such that light from the light emitting element is transmitted through the lens part of the lens.

10. The method of manufacturing a light emitting device according to claim 9, wherein the disposing the lens comprises, mounting a light emitting element on a substrate, and bonding the one or more flange parts to the substrate.

11. The method of manufacturing a light emitting device according to claim 9, wherein in the disposing the lens, the lens is disposed such that the connection part and the lens part are spaced apart from the light emitting element.

12. The method of manufacturing a light emitting device according to claim 11, wherein in the disposing the lens, the lens is disposed such that a distance between the lens part and the light emitting element is smaller than a distance between the connection part and the light emitting element.

13. A method of manufacturing a light emitting device, the method comprising: providing a lens using a method according to claim 2, and disposing the lens such that light from the light emitting element is transmitted through the lens part of the lens.

14. The method of manufacturing a light emitting device according to claim 13, wherein the disposing the lens comprises, mounting a light emitting element on a substrate, and bonding the one or more flange parts to the substrate.
 15. The method of manufacturing a light emitting device according to claim 13, wherein in the disposing the lens, the lens is disposed such that the connection part and the lens part are spaced apart from the light emitting element.
 16. The method of manufacturing a light emitting device according to claim 15, wherein in the disposing the lens, the lens is disposed such that a distance between the lens part and the light emitting element is smaller than a distance between the connection part and the light emitting element.
 17. A method of manufacturing a light emitting device, the method comprising: providing a lens using a method according to claim 3, and disposing the lens such that light from the light emitting element is transmitted through the lens part of the lens.
 18. The method of manufacturing a light emitting device according to claim 17, wherein the disposing the lens comprises, mounting a light emitting element on a substrate, and bonding the one or more flange parts to the substrate.
 19. The method of manufacturing a light emitting device according to claim 17, wherein in the disposing the lens, the lens is disposed such that the connection part and the lens part are spaced apart from the light emitting element.
 20. The method of manufacturing a light emitting device according to claim 19, wherein in the disposing the lens, the lens is disposed such that a distance between the lens part and the light emitting element is smaller than a distance between the connection part and the light emitting element.
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